Problem Set #3

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Question 1

Assume that individuals have the same utility function over consumption and labor given $U(c, l) = c - \frac{l^2}{2}$, where c represents consumption and l represents hours of labor. Assume also that the only income that individuals have is from labor income, that the hourly wage rate is given by $w = 100$ and the tax rate is given by τ.

(a) **Write the budget constraint faced by the individual.**

(b) **Does a tax rate $\tau = 100\%$ maximize government revenue in this situation? Explain why or why not.**

(c) **Find the labor supply function.**

(d) **Find the tax rate that maximizes government revenue.**

Assume that instead the government imposes the following tax schedule:

- **Any gross income between 1 and 5000 is taxed at 10%**
- **Any gross income between 5001 and 10000 is taxed at 20%**
- **Any gross income between 10001 and 15000 is taxed at 30%**

(e) **Graph the average tax rate and the marginal tax rate as a function of gross income.**

(f) **Is the tax schedule progressive or regressive? Explain.**

(g) **Assuming that the individual has the same utility function as before, find the number of hours supplied in equilibrium and the marginal tax rate it corresponds to.**

(h) **Calculate the average tax rate of the individual.**

(a) The individual has no endowment, the only income they get is their labour income. Clearly, the total amount of income earned from labour is wl. However, their consumption is not equal to this since they have to pay a tax at the rate of τ. So, they end up paying $wl\tau$ as tax and hence, since net income has to be at least as much as the consumption. So their budget constraint is:

$$c \leq wl(1-\tau)$$

In particular, for this case:

$$\boldsymbol{c \leq 100l(1-\tau)}$$

(b) When $\tau = 100\%$, $c = wl(1-1) = 0$. Hence, people are not better off by working and they have no incentive to work. This results in nobody doing their jobs and hence no tax revenue for the government. So clearly, $\tau = 100\%$ does **not** maximise government revenue.

(c)

$$max_{c,l} U(c,l) = c - \frac{l^2}{2}$$

such that

$$c \leq wl(1-\tau)$$

Clearly, utility will be maximised when

$$c = wl(1-\tau)$$

Plugging this into the objective function, we get

$$U(c,l) = wl(1-\tau) - \frac{l^2}{2}$$

Differentiating w.r.t l:

$$\frac{dU}{dl} = w(1-\tau) - \frac{2l}{2}$$

$$\implies \frac{dU}{dl} = w(1-\tau) - l$$

First order condition:

$$w(1-\tau) - l = 0$$

$$\implies l = w(1-\tau)$$

Checking second order condition to ensure maxima:

$$\frac{d^2U}{dl^2} = -1$$

$$\implies \frac{d^2U}{dl^2} < 0$$

$\implies$ Hence, the labour supply function is $\boldsymbol{l = w(1-\tau)}$ which is equal to $\boldsymbol{100(1-\tau)}$ in this case.

(d)

$$max_{\tau} T = \tau w l$$

We know that

$$l = w(1-\tau)$$

Therefore

$$\tau w l = \tau w^2 (1-\tau)$$

Differentiating w.r.t τ:

$$\frac{dT}{d\tau} = w^2(1-\tau) - \tau w^2$$

$$\implies \frac{dT}{d\tau} = w^2 - 2\tau w^2$$

First order condition:

$$w^2 - 2\tau w^2 = 0$$

$$\implies w^2 = 2\tau w^2$$

Since $w \neq 0$

$$2\tau = 1$$

$$\implies \tau = 0.5 = 50\%$$

$\implies$ Hence, the tax rate that maximises government revenue is **50**%.

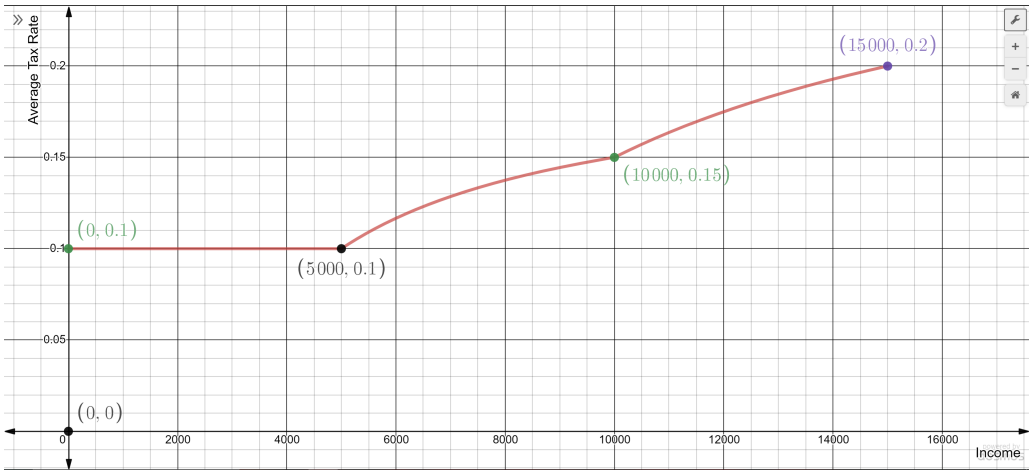
(e) I have used the website *www.desmos.com* for my graphing purposes.

Average Tax Rate

The average tax rate τ can be modelled as a function of the gross income $x = wl$ as:

$$\tau = \begin{cases} 0.1 & 0 \leq x \leq 5000 \\ \frac{500+0.2(x-5000)}{x} & 5001 \leq x \leq 10000 \\ \frac{500+1000+0.3(x-10000)}{x} & 10001 \leq x \leq 15000 \end{cases}$$

The graph for the same is drawn as follows:



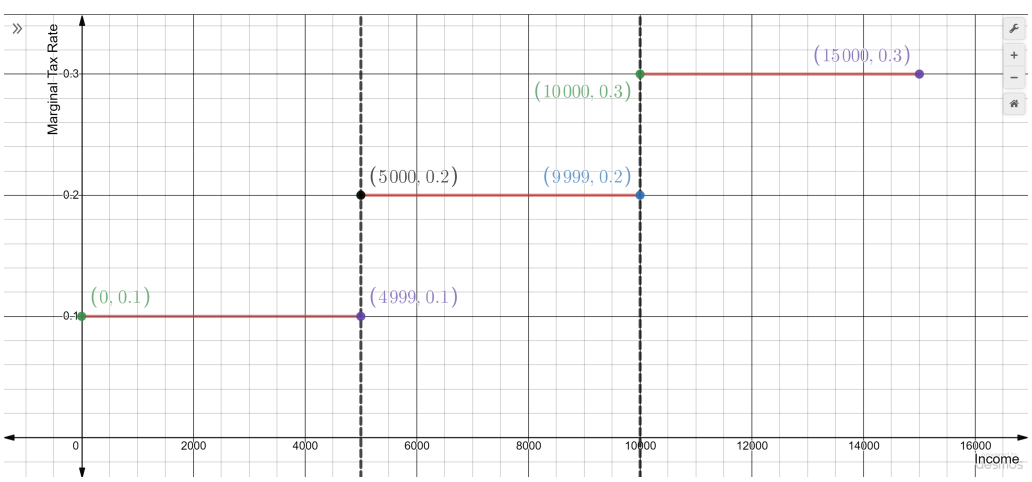
Note that the lines joining the two extremes of the second and third slabs are not straight, since the average tax rate does **not** increase at a constant rate and is a function of x, as shown earlier.

Marginal Tax Rate

The marginal tax rate t' is fairly simple to graph, as compared to the average tax rate. It can be modelled as a function of the gross income x as:

$$t' = \begin{cases} 0.1 & 0 \leq x \leq 4999 \\ 0.2 & 500 \leq x \leq 9999 \\ 0.3 & 10000 \leq x \leq 15000 \end{cases}$$

The graph for the same is drawn as follows:

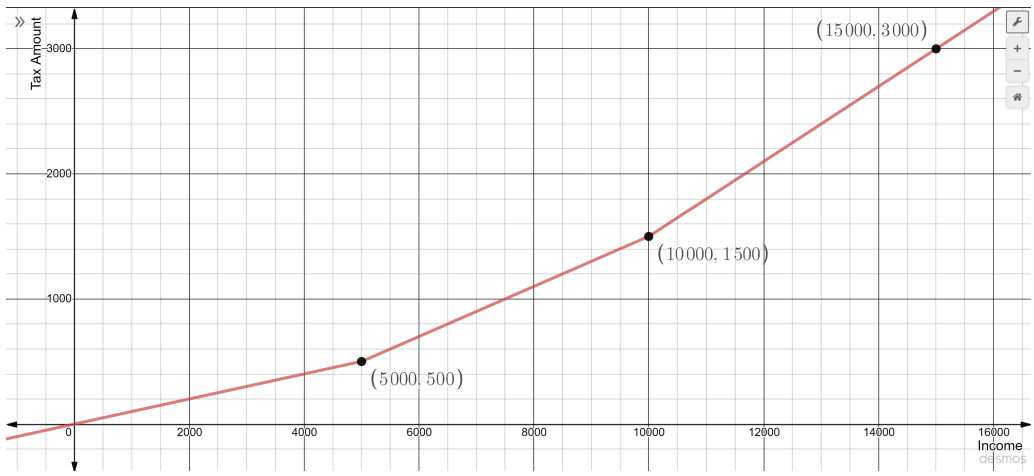


Tax Amount

The total tax amount of tax T paid can be modelled as a function of the gross income x as:

$$T = 0.1x + 0.1(\max(0, (x - 5000))) + 0.1(\max(0, (x - 10000)))$$

The graph for the same is drawn as follows:



(f) A progressive tax is defined as a tax whose rate increases as the payer's income increases. That is, individuals who earn high incomes have a greater proportion of their incomes taken to pay the tax. This is true for the given schedule as the richer individuals are taxed a higher percentage of their income, hence the given schedule is **progressive**.

(g) This question seems to have a circular dependency and is hence, harder to solve. The average tax rate, as we have shown, depends on the gross income, which clearly depends upon the labour supply, which in turn depends on the average tax rate.

Let us take a look again at the average tax function we modelled:

$$\tau = \begin{cases} 0.1 & 0 \leq x \leq 5000 \\ \frac{500+0.2(x-5000)}{x} & 5000 \leq x \leq 10000 \\ \frac{500+1000+0.3(x-10000)}{x} & 10000 \leq x \leq 15000 \end{cases}$$

Now, clearly the gross income x can be replaced with wl

$$\tau = \begin{cases} 0.1 & 0 \leq wl \leq 5000 \\ \frac{500+0.2(wl-5000)}{wl} & 5000 \leq wl \leq 10000 \\ \frac{500+1000+0.3(wl-10000)}{wl} & 10000 \leq wl \leq 15000 \end{cases}$$

According to the question, $w = 100$. Substituting it in the average tax function, we get:

$$\tau = \begin{cases} 0.1 & 0 \leq 100l \leq 5000 \\ \frac{500+0.2(100l-5000)}{100l} & 5000 \leq 100l \leq 10000 \\ \frac{500+1000+0.3(100l-10000)}{100l} & 10000 \leq 100l \leq 15000 \end{cases}$$

Which can be rewritten as:

$$\tau = \begin{cases} 0.1 & 0 \leq l \leq 50 \\ \frac{5+0.2(l-50)}{l} & 50 \leq l \leq 100 \\ \frac{15+0.3(l-100)}{l} & 100 \leq l \leq 150 \end{cases}$$

The labour supply equation, as calculated earlier, is:

$$l = w(1-\tau)$$

which can be written as:

$$l = 100(1-\tau)$$

where τ is **not** the marginal tax rate, but the average tax rate, since we got this function from maximising the utility function, which takes into account the **average tax rate**.

Now, to get to the equilibrium, let us consider any random value of τ to start with, since it doesn't matter what we pick initially. Let us set $\boldsymbol{\tau = 0.2}$, then we see that

$$l = 100(1-0.2)$$

$$\implies l = 80$$

Using the simplified average tax function described above, we get

$$\tau = \frac{5+0.2(80-50)}{80}$$

$$\implies t = \frac{5+6}{80}$$

$$\implies \boldsymbol{t = 0.1375}$$

Now, this is clearly not equal to the value of 0.2 that we started with. So we have not yet arrived at an equilibrium.

Now, $\boldsymbol{\tau = 0.1375}$. So we see:

$$l = 100(1-0.1375)$$

$$\implies l = 86.25$$

Using the simplified average tax function described above, we get

$$\tau = \frac{5+0.2(86.25-50)}{86.25}$$

$$\implies \tau = \frac{5+7.25}{86.25}$$

$$\implies \boldsymbol{\tau \approx 0.142}$$

Now, this is clearly not equal to the value of 0.1375 that we started with. So we have not yet arrived at an equilibrium.

Now, $\boldsymbol{\tau = 0.142}$. So we see:

$$l = 100(1 - 0.142)$$
$$\implies l = 85.8$$

Using the simplified average tax function described above, we get

$$\tau = \frac{5 + 0.2(85.8 - 50)}{85.8}$$
$$\implies \tau = \frac{5 + 7.16}{85.8}$$
$$\implies \boldsymbol{\tau = 0.1417}$$

We are pretty close, but let's go for one more iteration, to get a precise answer.

Now, $\boldsymbol{\tau = 0.1417}$. So we see:

$$l = 100(1 - 0.1417)$$
$$\implies l = 85.83$$

Using the simplified average tax function described above, we get

$$\tau = \frac{5 + 0.2(85.83 - 50)}{85.83}$$
$$\implies \tau = \frac{5 + 7.166}{85.83}$$
$$\implies \boldsymbol{\tau = 0.1417}$$

Now, we see that we have arrived at an equilibrium

$\implies$ Hence, at the equilibrium, the number of hours supplied is ≈ 85.83. But since wages are paid per hour and I am not sure whether fractions of hours are allowed, we can say that the number of hours supplied at the equilibrium are **86**.

(h) The average tax rate that the individual faces at equilibrium is

$$\tau = \frac{5 + 0.2(86 - 50)}{86}$$
$$\tau = \frac{5 + 7.2}{86}$$
$$\implies \boldsymbol{\tau \approx 14.19\%}$$